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Technologies in Fog Computing

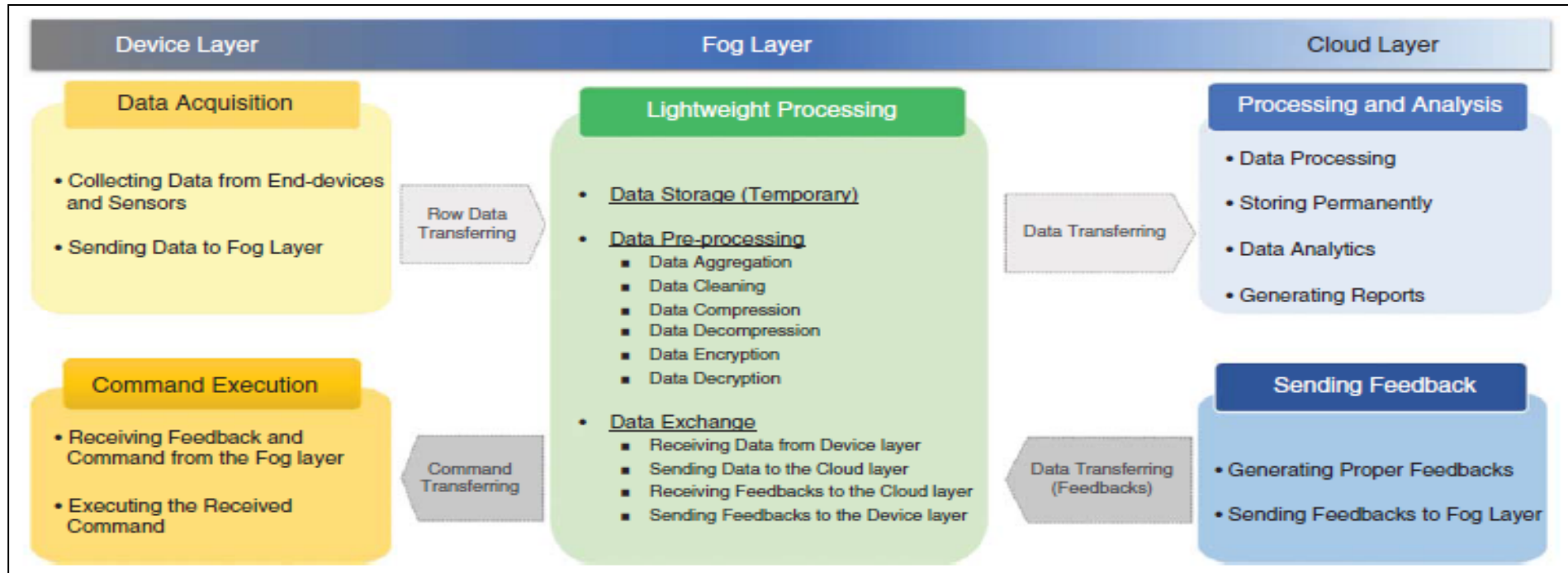
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FOG DATA MANAGEMENT

- Data life cycle and fog data characteristics as the essential concepts of fog data Management

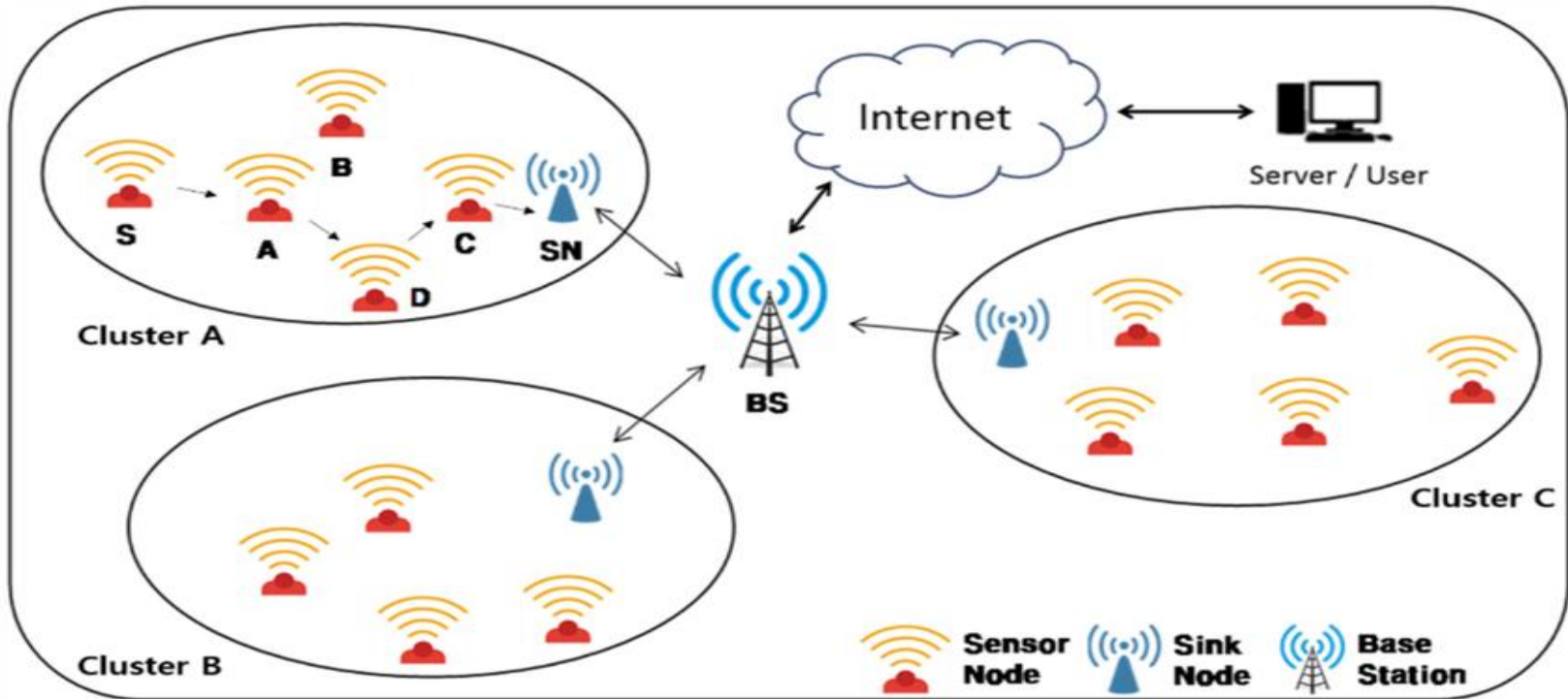
A. Fog Data Life Cycle

- The fog data life cycle consists of 5 main steps *data acquisition, lightweight processing, processing and analysis, study feedback, and command execution*



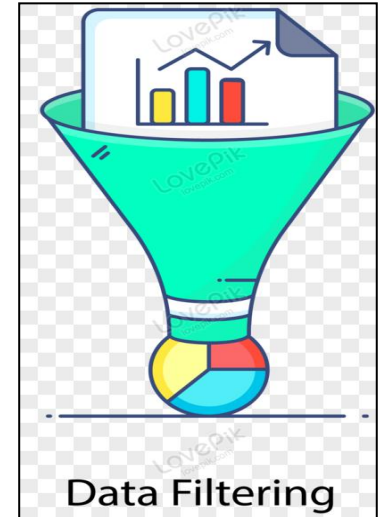
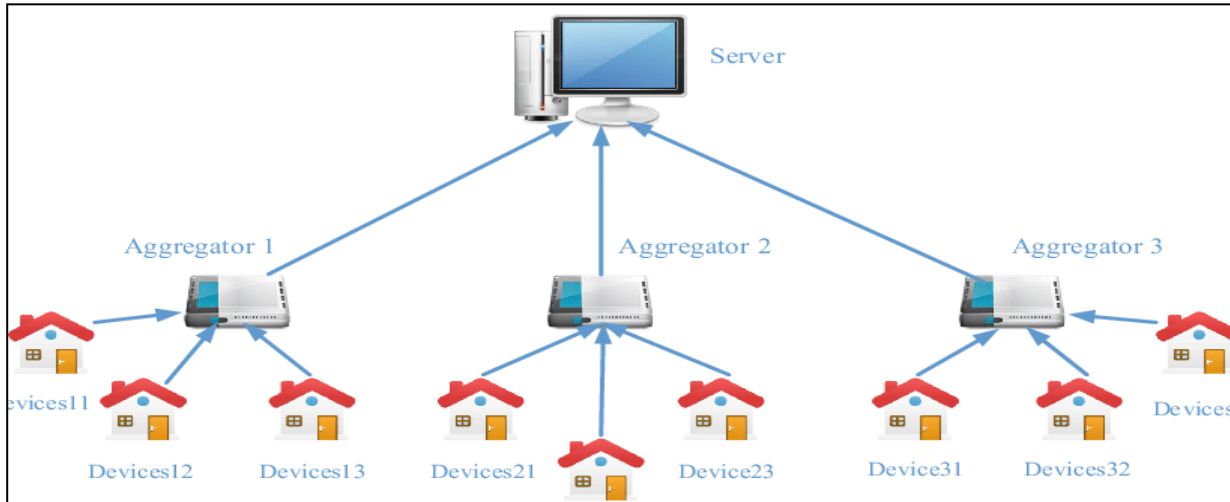
i) Data Acquisition

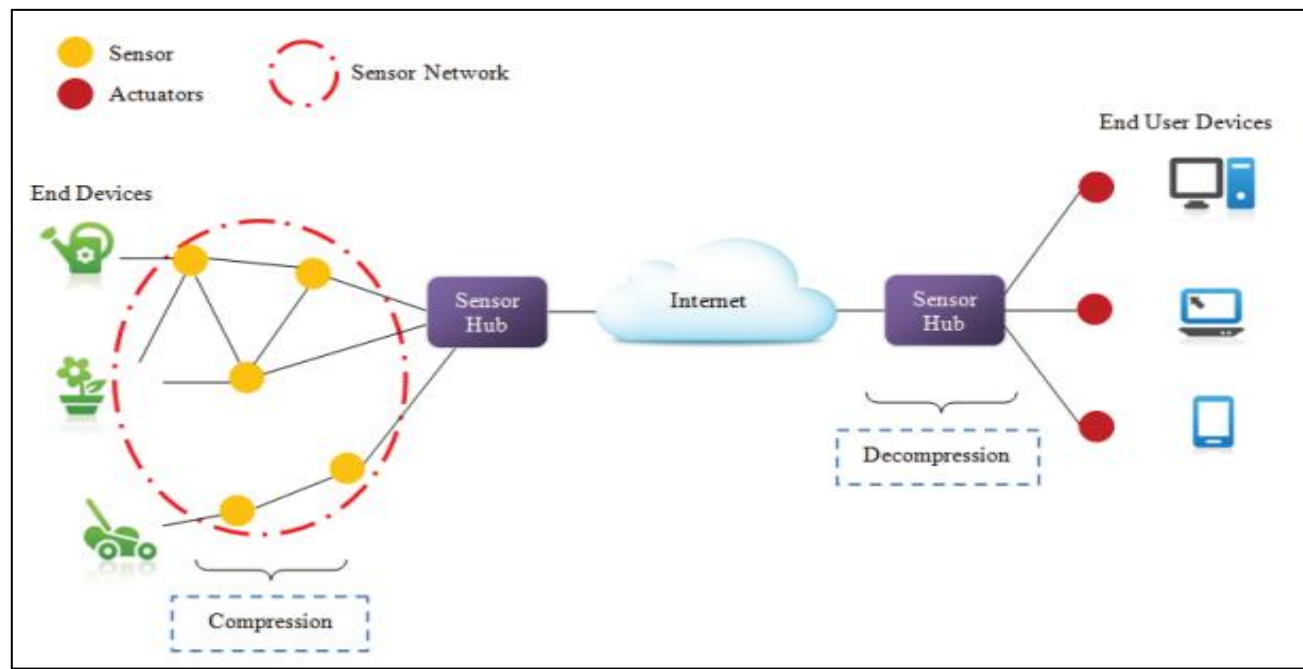
- Data from different types of end devices should be acquired.
- It must be sent to upper layers.
- To this end, a **sink node** or **local gateway node** may exist to gather data, or sensors can send data directly to the fog.



ii) Lightweight Processing

- This step provides **lightweight data manipulation** and **local data processing** on the collected data.
- Lightweight processing may include
 - *data aggregation*
 - *data filtering, elimination of unnecessary or repetitive data*
 - *data cleaning*
 - *compression/ decompression*
 - *some lightweight data analysis and pattern extraction*

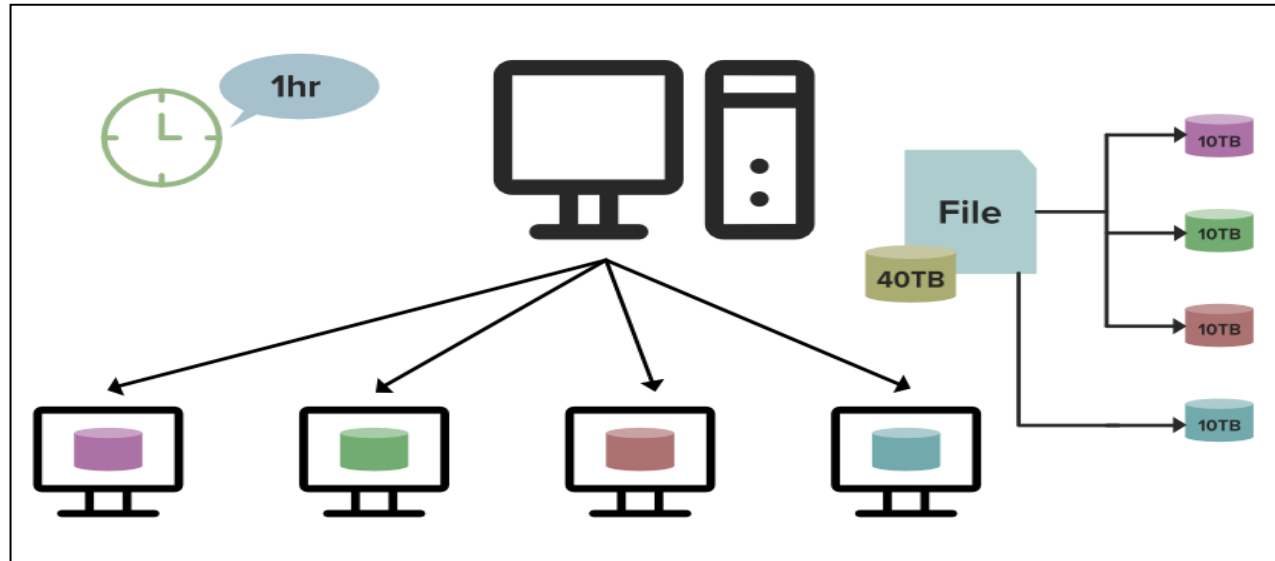




- The aggregated data will be transferred to the cloud via the network.
- In addition, the **feedback as response data** should be transferred to the device.
- Also, as the feedback is received from cloud layer and sent to the device layer, there may be a need for data decompression, data decryption, doing some format change on the received data, etc.

iii) Processing and Analysis

- Received data may be **stored permanently in the cloud layer**, and it is processed based on predefined requirements.
- The application users may access data to get reports or data analysis.
- Different types of analysis on stored data may be applied to obtain valuable information and knowledge, and these types of processing and analysis are almost in the scale of **big data**.
- They need big data platforms and technologies such as **HDFS for storage** and **map-reduce for processing**.



iv) Sending Feedback

- Based on data processing and analyzing, feedbacks such as proper commands or decisions are generated and sent to the fog layer.

v) Command Execution

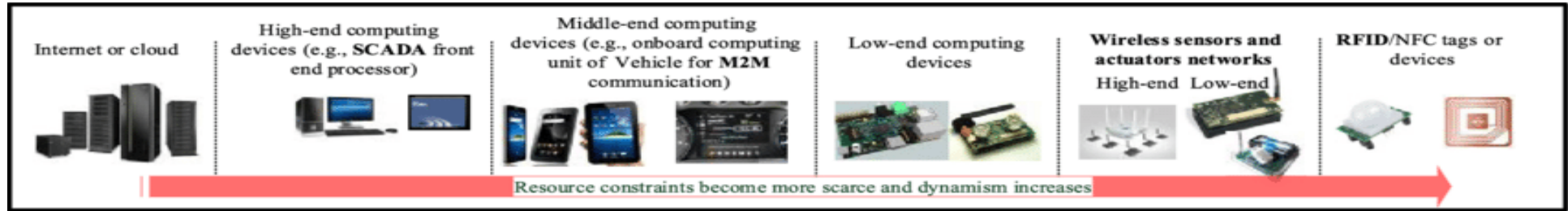
- **Actuators** must run the proper action based on the received data.
- In this way, proper feedback and responses are applied to the environment.

B. Data Characteristics

- Reviewing data characteristics helps set quality and integration standards while addressing challenges in data management.
- **Data quality** refers to how much data characteristics are **suitable** and can **meet the terms** with **consumer requirements**.
- The main IoT data characteristics are as follow:
 - **Uncertainties**
 - **Errors**
 - **Noisy data**
 - **Voluminous and distributed**
 - **Gradual Changes**
 - **Continuous correlation**
 - **Patterns**
 - **Memory-Based**

- **Data quality dimensions** reviewed are
 - *Accuracy*
 - *Confidence*
 - *Completeness*
 - *Data volume*
 - *Timeliness*
- Three other additional data quality dimensions are
 - *Ease of access*
 - *Access security*
 - *Interpretability*
- IoT data characteristics were categorized into three categories:
 - *Data generation*
 - *Data quality*
 - *Data interoperability*
- The **IoT data quality** characteristics include
 - *Uncertainty*
 - *Redundancy*
 - *Ambiguity*
 - *Inconsistency*

- Management of main IoT data characteristics and the related issues can be done in the fog data management process to fulfill the requirements.
- In the following, these characteristics are reviewed:
- **Heterogeneity.**
 - Distributed heterogeneous end devices generate data in different formats.



- Generated data may be diversely varying in terms of **structure** or **format**.
- **Inaccuracy.**
 - Inaccuracy or uncertainties of the sensed data refer to the **sensing precision, accuracy, or misreading of data.**
- **Inconsistency.**
 - **Low precision** or **misreading** in the sensed data may cause inconsistency in the gathered data.

- **Weak semantics**

- The collected raw data that may be heterogeneous in terms of **data formats**, **data structure**, **data source**, etc. must be processed and managed.
- Using the concepts of **semantic web and injecting** some information and extra data to the raw data make the data readable and understandable for machines.
- Most of the collected data from the environment has weak semantics

- **Velocity**

- Data generation rates and sampling frequencies are varying in different types of end devices.

- **Redundancy.**

- Repetitive data that are sent by one or more end devices lead to redundancy in the collected data.

- **Scalability.**

- Large numbers of end devices and high data sampling rate that may exist in different scenarios may lead to generation of a huge amount of data.

C. Data Pre-Processing and Analytics

i) Data Cleaning

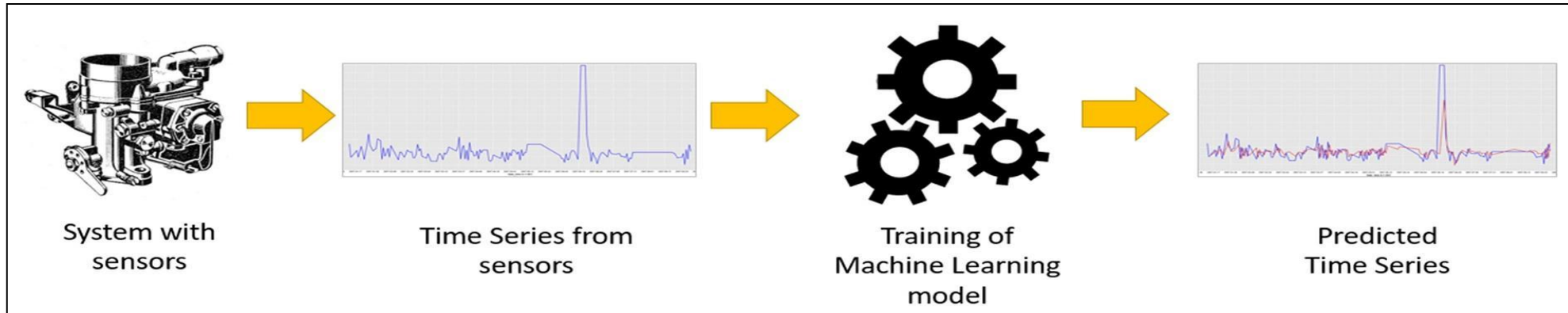
- Cleaning mechanisms can be applied on the collected data in fog layers to **reduce the effect of dirty and unreliable data**, and to **increase the quality** of them.
- Data cleaning approaches can be divided in two categories:
 - **Declarative Data Cleaning**
 - **Model-based Data Cleaning**

Declarative Data Cleaning

- High-level declarative queries such as **CQL** (Continuous Query Language) are used to **define the sensor values constraints**.
- The user can express the queries and control the system easily via the provided interface.
- Extensible sensor stream processing (ESP), is a declarative-based and pipelined framework for sensor data cleaning for use in pervasive applications.

Model-based Data Cleaning

- Anomalies are detected by **comparison of actual values** with the **predicted values** that are resulted as the most probable values based on selected models.
- The model-based approaches also have subcategories such as
 - **Regression models**
 - **Polynomial regression**
 - **Chebyshev regression**
 - **Probabilistic models**
 - **Kalman filters**
 - **Outlier detection models**



ii) Data Fusion

- Data fusion refers to the **elimination of redundant** and **ambiguous data** and **integration of data**, and can be done in the fog layer as one of the data management tasks to increase the accuracy and efficiency.
- Data fusion models can be categorized into three particular categories:
 - **Data-based model**
 - **Activity-based model**
 - **Role-based model**
- Khaleghi et al. categorizes **data fusion frameworks** into four classes based on data-related aspects:

(i) Imperfect data fusion framework

- Related to data imperfection, which is one the main data challenges and caused by impreciseness, incompleteness, vagueness, or uncertainty

(ii) Correlated data fusion framework

- Related to dependency of data.

(iii) Inconsistent data fusion framework

- Conflict of data

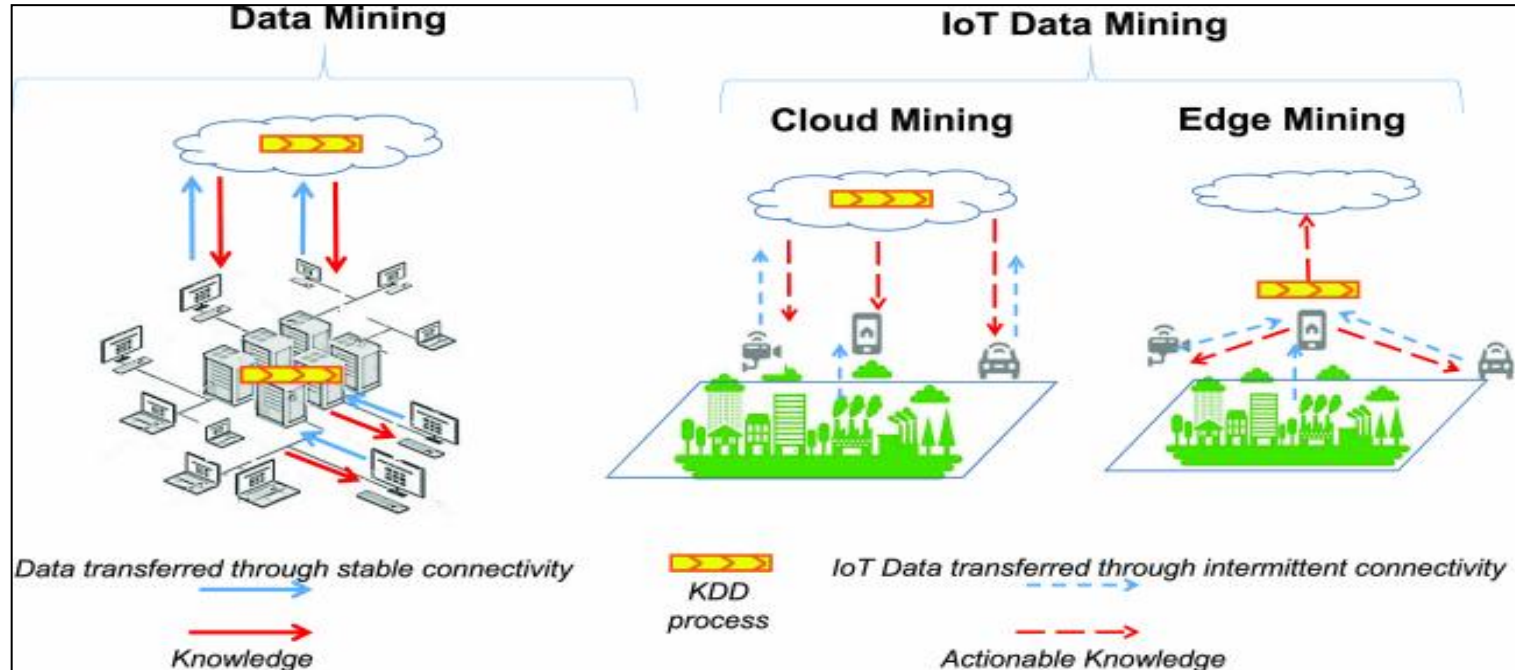
(iv) Disparate data fusion framework

- Diversity on data



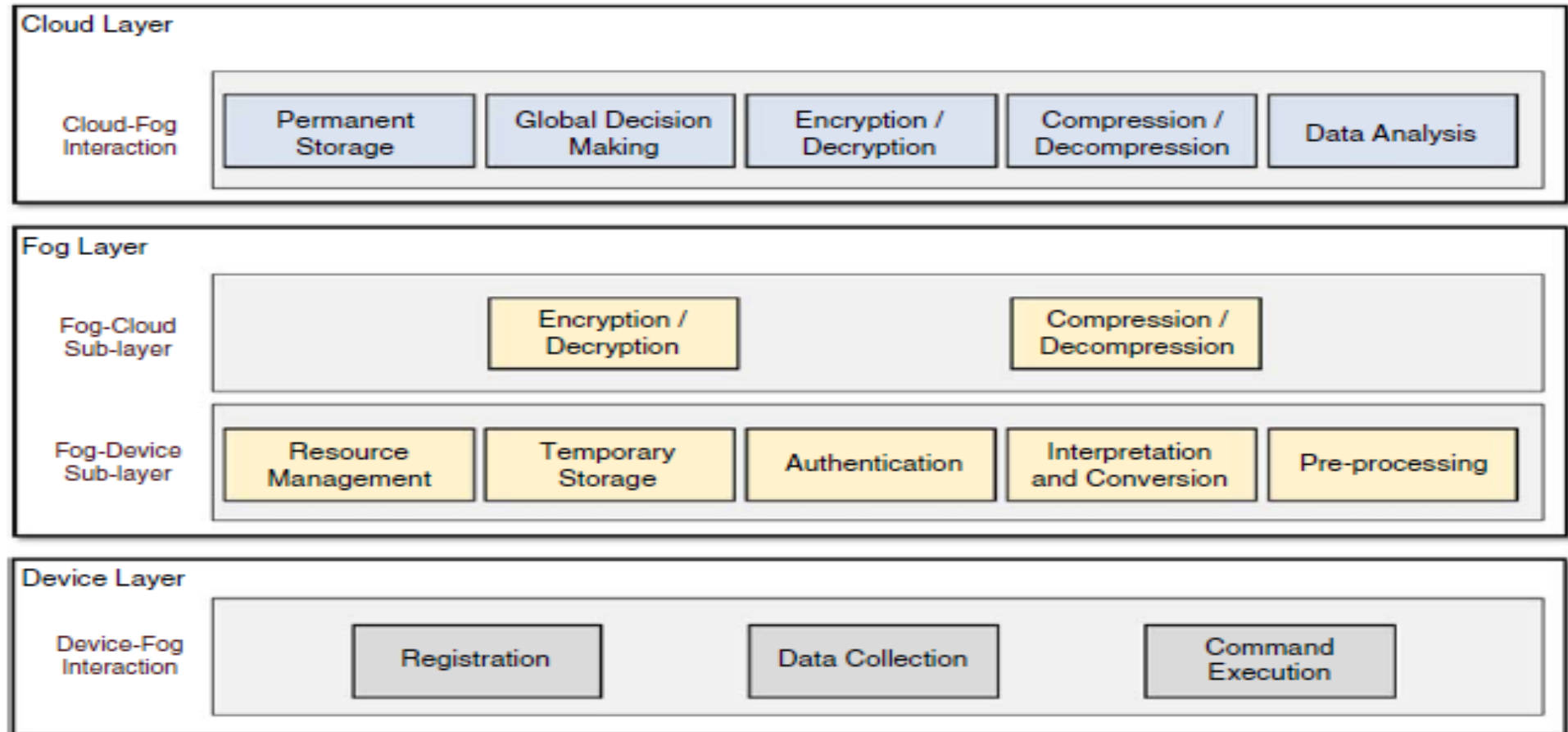
iii) Edge Mining

- Fog computing can be effective for local analytics and stream processing to reduce the volume of data.
- Edge mining refers to **utilize mining approaches on raw data** that are produced by **devices in the edge of the network** (fog layer).
- The size of the transferred data will be reduced and better energy savings can be achieved.



D. Proposed Architecture

- A conceptual architecture based on the three-layer model to handle data management issues is shown below



1. Device Layer

- Modules of the device layer are registration module, data collection module, and command execution module.
- **Registration.**
 - **Physical devices** can join to the network or leave it dynamically via this module.
 - Registration is necessary for **sending and receiving messages**.
 - Registration requests as an initial message should be sent by the device to the fog layer.
 - Through the registration process, devices get a **unique ID and key** that should be attached in the messages for authentication process.
- **Data Collection/Command Execution.**
 - **Registered sensors collect data** to transfer to the fog layer.
 - **Actuators** are responsible for **running the received commands** from the fog layer by this module.

2. Fog Layer

- ***Resource Management***

- The fog layer receives the **join requests** that are sent by the device layer.
- The resource management module **queries the list of devices** and **adds the device specifications** to the list in case of absence or deactivation of device, and it will be added.
- Based on **fog applications policy**, registered devices that are not sending message for a predefined period may be deactivated.

- ***Temporary Storage***

- Temporary storage can be a module to **store some of the incoming data** or **intermediate computation results**, for further processing in a database.
- It also stores **specifications of registered devices** and their **IDs** and **keys** for authentication process.

- ***Authentication.***
 - The authentication module **searches the list of registered devices** on **temporary storage** to find related key and ID to authenticate the incoming messages.
- ***Interpretation and Conversion.***
 - The communication of devices and fog may be receiving data, as well as there may be possible heterogeneity in terms of communication types between devices and the fog layer, so data may be sent via different technologies such as Wi-Fi, Bluetooth, ZigBee, RFID, or etc.
 - Therefore, Interpretation and conversion provides different **protocols** and **conversion methods**.
- ***Pre-Processing***
 - **Storing** the received data in the temporary storage, **preparing** and **aggregating** them are the responsibilities for the pre-processing module.

- Some kinds of data processing may be needed on the
 - **received data**, such as data cleaning, data fusion, edge mining,
 - **quality improvement of received signals** by data filtering or denoising them
 - **decision making in case of emergency situations** by checking and comparing the received or collected data against the predefined threshold and conditions
- ***Encryption/Decryption.***
 - To improve data privacy and protect sensitive data in the message, encryption and decryption algorithms are provided in this module.
- ***Compression/Decompression.***
 - Compression techniques can be applied on the **packet size of data** to reduce overhead of networks.

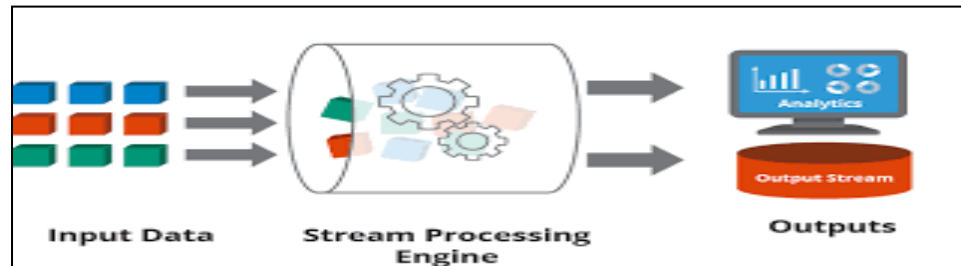
3. Cloud Layer

- **Permanent Storage**

- This module receives data from **different fog zones** and **stores the data** in permanent storage.
- Depending on the fog application type, the size of permanent storage might vary from **gigabytes** to even **petabytes**.
- Big data technologies are applied in this layer to store data.

- **Global Decision-Making**

- Received data are **processed to send proper feedback** to the lower layers and to store the needed and useful data in permanent storage.
- In case of receiving/sending **compressed** or **encrypted** messages from fog layer, **decompression** or **decryption** units are used, respectively.
- Data in motion and data stream processing are the other issues raised in IoT and fog computing for some kinds of applications.



- ***Encryption/Decryption, Compression/Decompression.***
 - For privacy and efficiency concerns and to support encryption and compression, these modules must be located in both the **cloud and fog layers**.
 - Encryption and compression approaches must be agreed by both of them.
- ***Data Analysis.***
 - Gathering of data will be very valuable when it leads to knowledge, so **data analysis, pattern recognition, and knowledge discovery from the heterogeneous data** that are collected from different devices are considered an important stage in the data life cycle.
 - This module is responsible for data analysis based on the application user requests for **providing reports** and **getting global analysis** on the gathered data.
 - Depending on the size of stored data in data storage, data analysis approaches and technologies may vary.
 - In the case of large-scale data storage, big data analytics technologies may be applied.

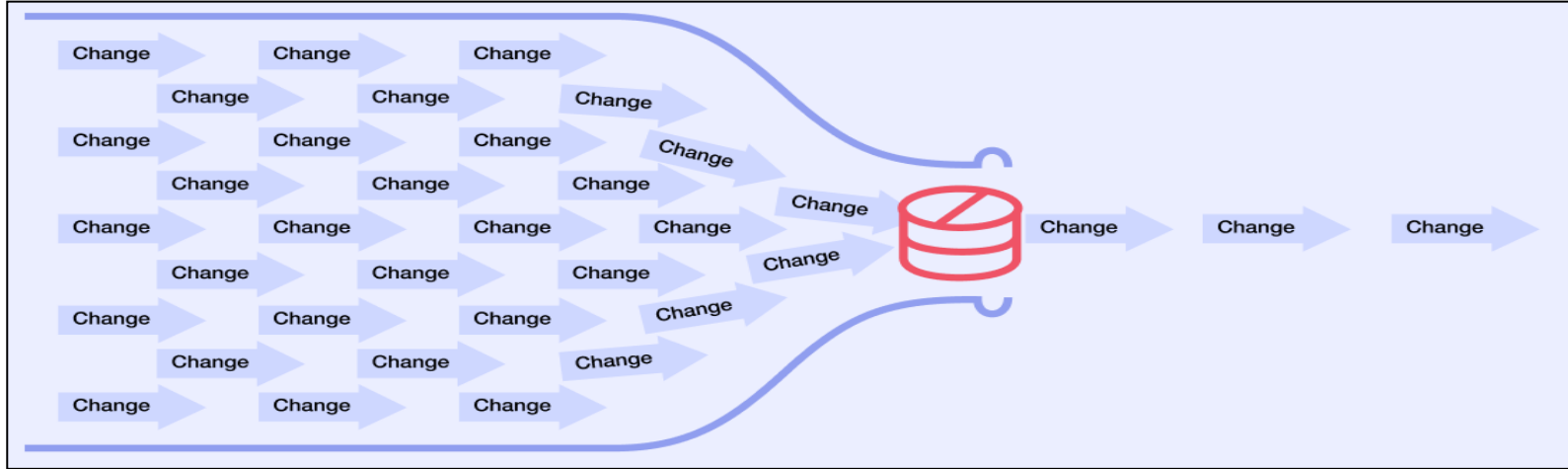
Machine Learning in Fog Computing

- A **cloud-only architecture** for IoT devices has several issues when data is sent to the cloud for processing and analysis:
 - i) **Network Traffic Congestion.**
 - By 2030, over 100 billion devices will be connected to the Internet.
 - If all data is processed in the cloud, it could cause network congestion and slow data transfer.



ii) Data Bottleneck.

- Storing and analyzing data only in the cloud can cause delays.
- This happens if the server is slow due to large data volume or other factors.

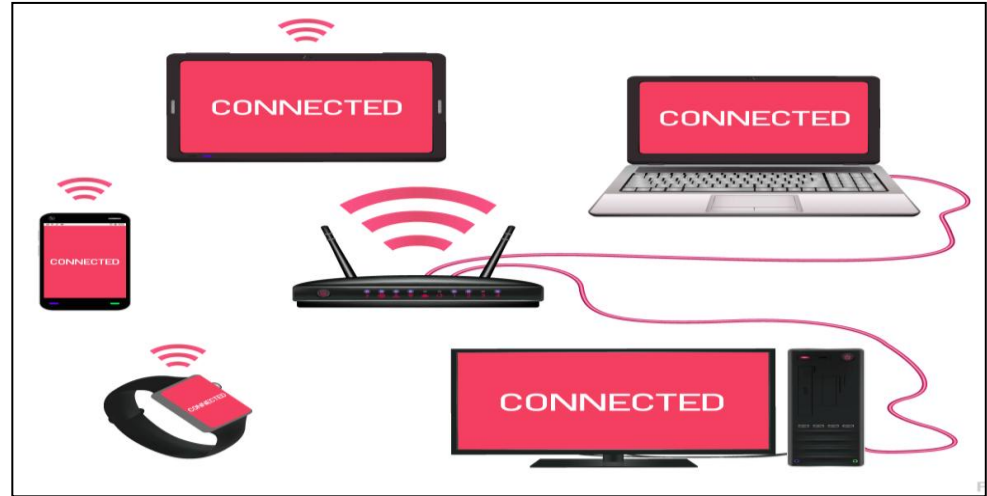


iii) Security Issues.

- As data moves through layers—from sensors to the cloud—it faces many vulnerabilities.
- Cloud security can handle common issues but may not protect individual sensors or edge nodes.

iv) Data Staleness.

- Data often *loses its value* if it is not analyzed quickly.
- Devices like security cameras, phones, cars, and ATMs generate data that need instant analysis during security or privacy issues.
- Fog computing solves this by bringing computing, storage, and decision-making closer to the data source.
- Key features of fog computing include low latency, location awareness, and both wired and wireless access.



There are numerous benefits as follow:

i) **Real-time Analytics.**

- As IoT adoption increases, the need for real-time analytics becomes more frequent, such as detecting intruders on security cameras or identifying fraudsters accessing accounts.



- Uploading and analyzing data in the cloud can be slow.
- Fog computing provides near-instant intelligence for such scenarios.

ii) Improved security.

- Fog computing, being closer to the edge, can apply device-specific security.
- It can also quickly decide whether to block access during a breach.

iii) Data thinning at the edge.

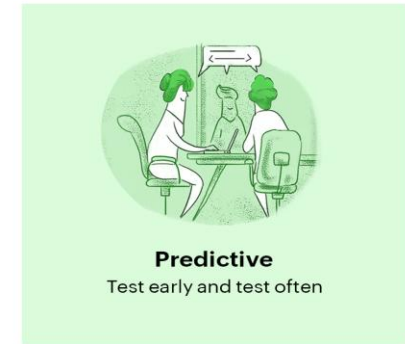
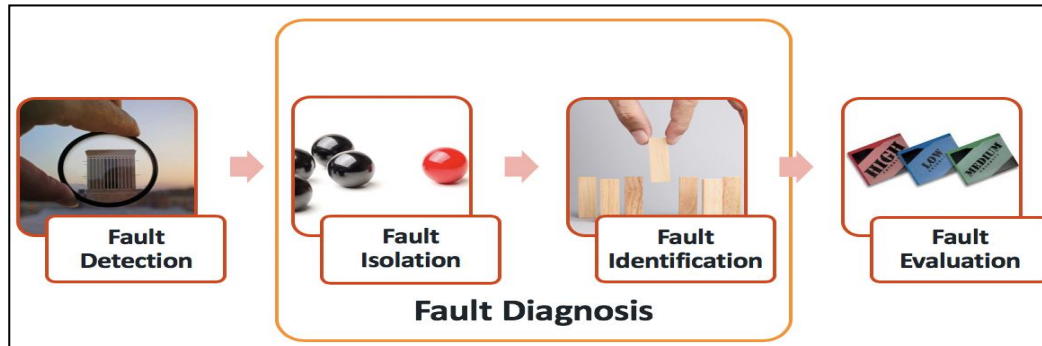
- Fog consumes the raw data and makes decisions or provides insights.
- It sends only relevant, consolidated information upward in the hierarchy.
- This dramatically reduces the amount of data that gets transmitted to a central data center.

iv) Cost Savings.

- Fog computing may have higher setup costs due to its distributed deployment, but its lower operational costs and long-term benefits make it worthwhile.

A. Machine Learning for Fog Computing and Security

- Fog computing enables near real-time analytics by using machine learning at fog nodes.
- In industries, this helps detect and isolate machine faults, reducing the Mean Time to Repair (MTTR) and improving system availability.
- Fog analytics can be reactive or predictive.
 - Nodes closer to the edge handle reactive analytics.
 - Nodes far away perform predictive analytics, as they need more computing power.
- In an n-tier architecture, computing power is highest in the cloud and decreases down the hierarchy.



- Machine-learning algorithms can be run at fog nodes that have the processing power to compute corresponding to the task at that layer .

Use case	ML algorithm
<i>Fog computing in industry</i> – Remote monitoring for oil & gas operations [57]	• Anomaly detection models • Predictive models • Optimization methods
<i>Fog computing in retail</i> – Retail customer behavior analysis [57]	• Statistical methods • Time series clustering
<i>Fog computing in self-driving cars</i> [57]	• Image processing • Anomaly detection • Reinforced learning

- Machine-learning models are created at the nodes near the cloud or in the cloud itself.
- The models could be downloaded to middle-tier nodes to help in execution.

B. Examples of Machine Learning in Fog Computing

i) ML in Fog Computing in Industry

- Traditional cloud analytics use past failure data to predict issues, but it's often too late to prevent them.
- Fog computing analyzes data locally in real time, helping prevent problems faster.

In **cloud** analytics, a factory machine detects a fault after it breaks down.

In **fog** computing, it detects unusual vibrations early and stops to prevent damage.



ii) ML in Fog Computing in Retail

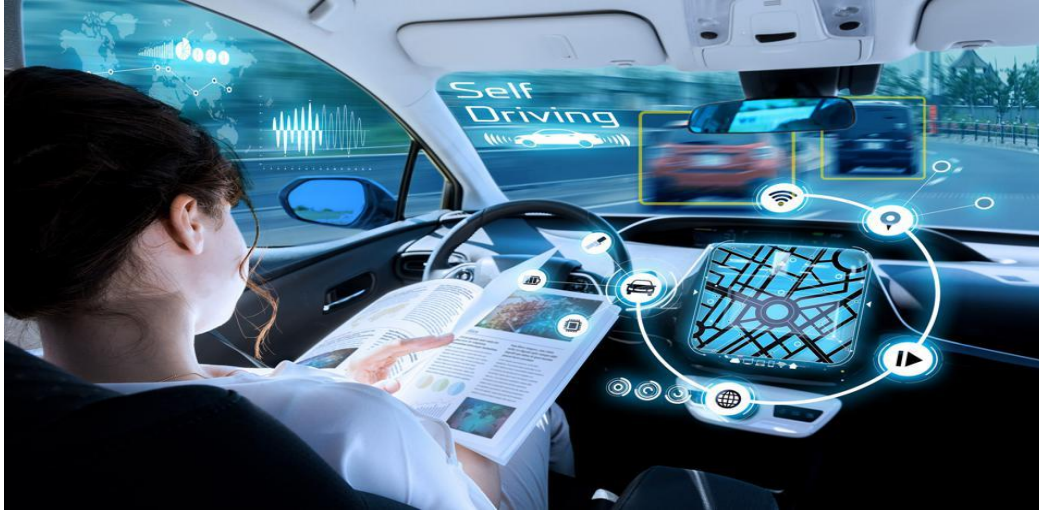
- Retail stores use analytics from *customer purchases* and *seasonal trends* for product placement, such as during Halloween or Christmas.
- With fog computing, stores can analyze local buying patterns in real time, helping them target merchandise better and enhance customer experience.

In **cloud** analytics, a store arranges Halloween items based on last year's sales.

In **fog** computing, it notices more candy sales today and quickly adds more near the checkout.

iii) Fog Computing for Self-Driving Cars

- Self-driving automobiles are excellent examples of fog computing, since a lot of computing and decision-making happens on the edge.



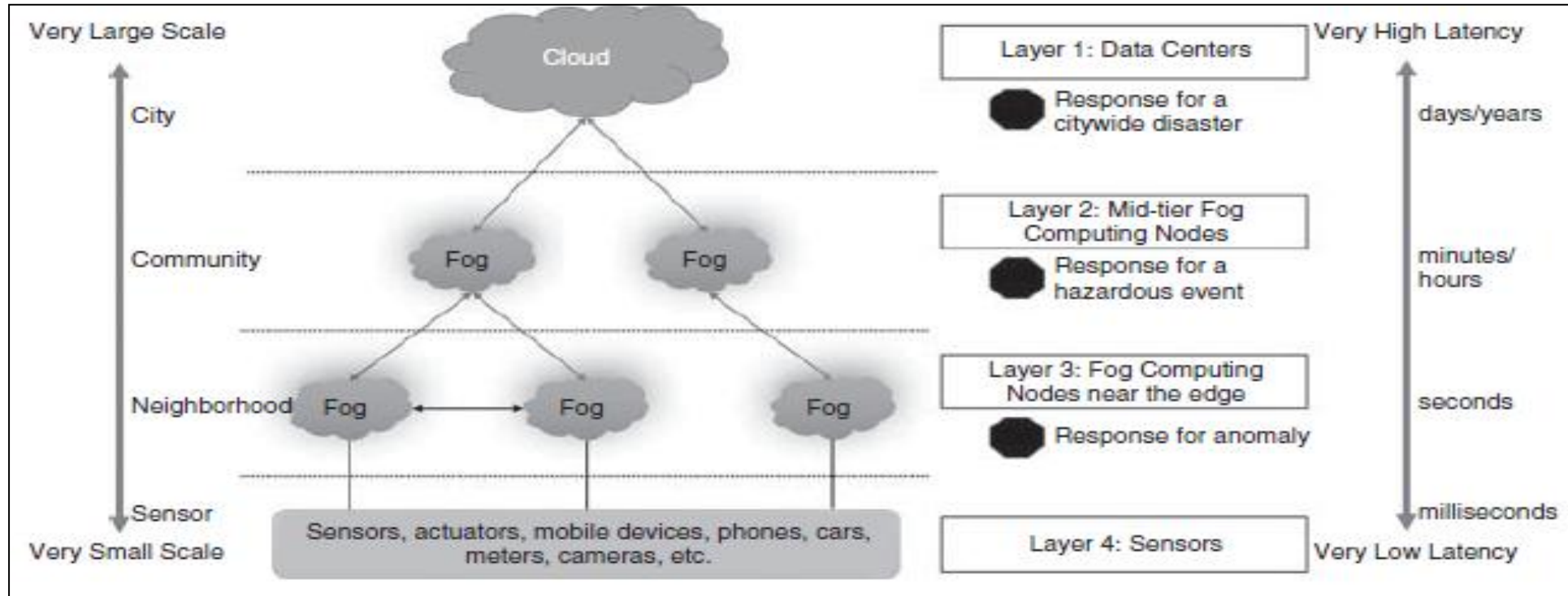
A self-driving car detects a pedestrian and stops instantly without waiting for cloud instructions.



- Each car transmits a lot of data for processing in the cloud.
- An N -tier model would make the system considerably more efficient.
- **Machine-learning algorithms** used are
 - ***ANN for image processing*** (A self-driving car uses ANN to identify road signs, recognize pedestrians)
 - ***Naive Bayes or similar algorithms for anomaly detection, reinforced learning.*** (Used Naïve Bayes to detect unusual driving patterns, like sudden lane changes.)

c. Machine Learning in Fog Computing Security

- Tang et al. presented a **hierarchical structure for fog computing architecture** to support the integration of massive number of infrastructure components and services in future **smart cities**.
- The architecture laid out in the paper is a **four-layer model**, with the first layer being the cloud and the last being the sensors.



- Layer 3 contains fog nodes that **get raw data from the sensors**.
 - The nodes at this layer perform two functions.
 - ✓ One **identifies potential threat patterns** on the incoming data streams from sensors using machine-learning algorithms.
 - ✓ The other **performs feature extraction** for reducing the amount of data to be sent upstream.
 - Algorithms like KNN, Naïve Bayes, random forests, or DBSCAN could be used to do anomaly detection.
- Layer 2 contains fog nodes that **get data from nodes below them**, and the data represent information from hundreds of sensors across locations.
 - In the paper, **HMM** (hidden Markov model) and **MAP** (maximum a posteriori) algorithms are used for **classification** and alert if there is a hazardous event.

- Machine-learning algorithms at different fog layers.

Layer	Disaster response	ML algorithm
<i>Layer 4 - Sensors</i>	None	None
<i>Layer 3 – Fog nodes for the neighborhood</i>	Response for anomaly	KNN, Naïve Bayes, random forest, DBSCAN
<i>Layer 2 – Fog nodes for the community</i>	Response for hazardous event	HMM, MAP [58] Regression, ANN, decision trees
<i>Layer 1 – Cloud</i>	Response for city-wide disaster, long-term forecasting	ANN, Deep learning, decision trees, reinforcement learning, Bayesian networks

Data Analytics in the Fog

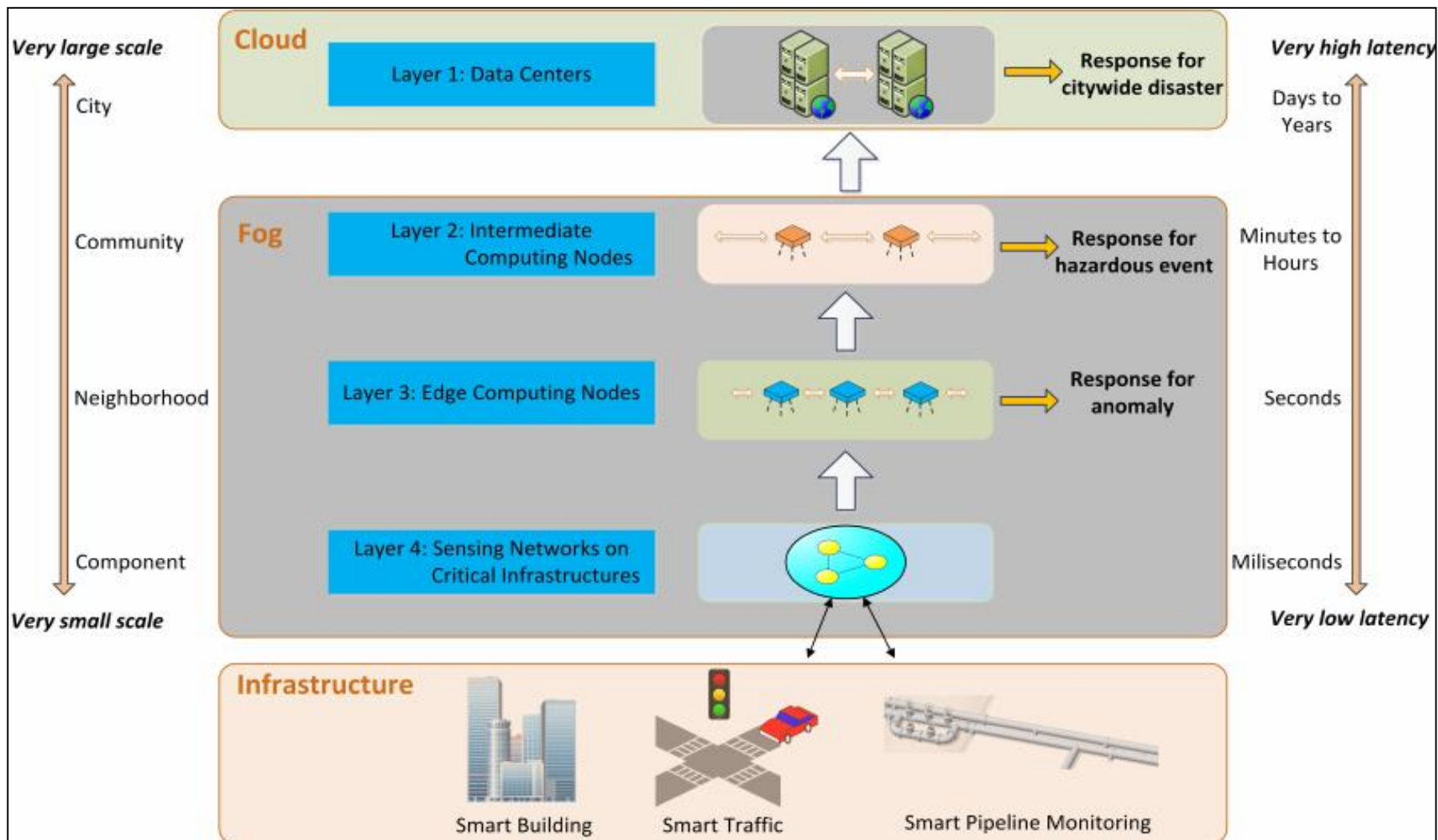
- Fog computing is a *virtual platform* that offers computing, storage, and networking services between end devices and cloud data centers, usually near the network edge.
- The fog has some distinctive characteristics that make it more appropriate for the applications requiring
 - **Low latency**
 - **Mobility support**
 - **Real-time interactions**
 - **Online analytics**
 - **Interplay with the cloud**
- Instead of sending all data to the cloud, an edge device or software solution may **perform a preliminary analysis** and **send a summary of the data** (or metadata) to the cloud.

i) Fog Analytics

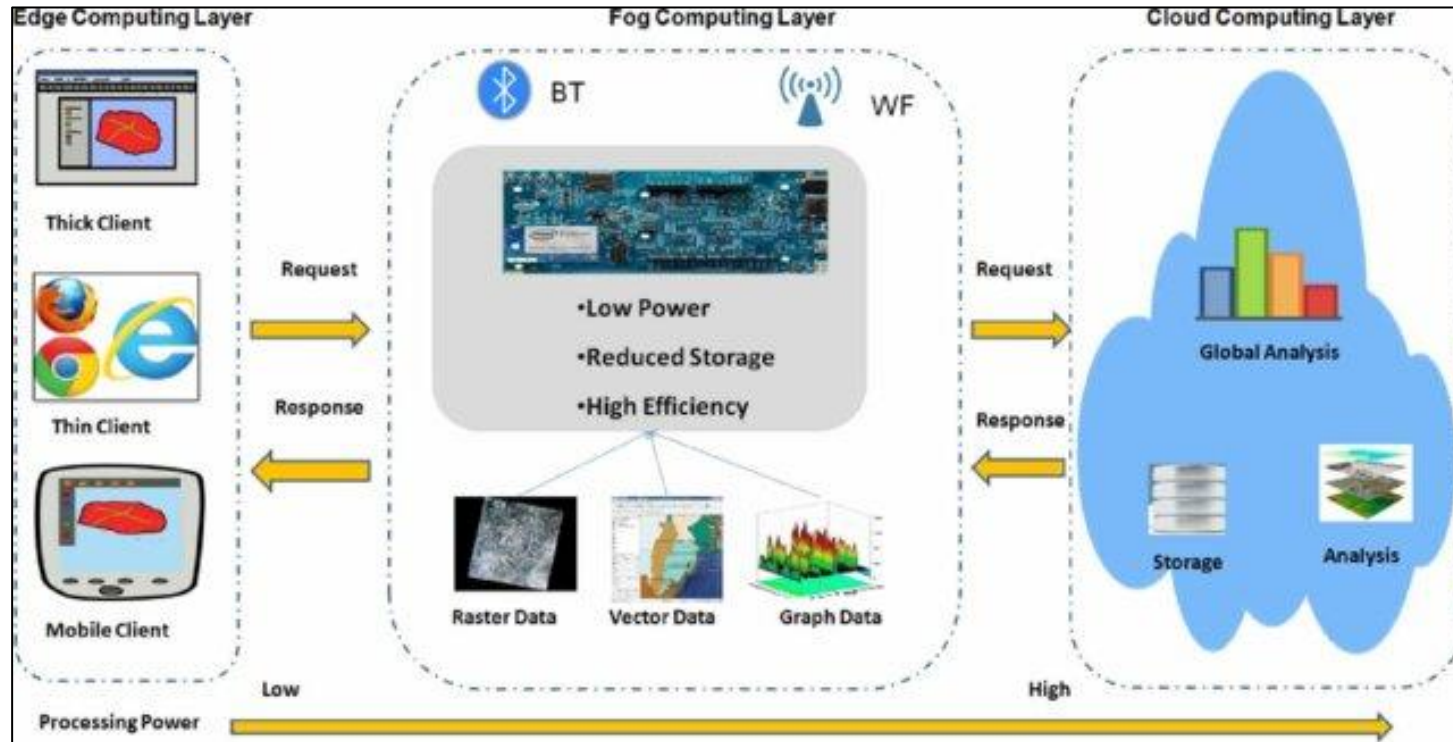
- Fog computing *processes data closer to the source* before it reaches the cloud, reducing communication time, cost, and the need for large storage.
- Fog offers **low latency** and **context awareness**, while the cloud provides **centralized global access**.
- Applications like big data analytics benefit from fog's speed and the cloud's scalability.
- The main function of the fog is
 - **to collect data from sensors and devices**
 - **process the data**
 - **filter the data**
 - **send the rest to the other parts** for
 - ✓ **Local storage**
 - ✓ **Visualization**
 - ✓ **Transmission to the cloud**

- The **cloud** provides local coverage and stores data for months or years, serving as the foundation for business intelligence analytics.
- Fog computing is still in early stages and present new challenges such as
 - ***Fog architecture***
 - ***Frameworks and standards***
 - ***Analytics models***
 - ***Storage and networking resource provisioning and scheduling, programming abstracts***
 - ***Models, and security and privacy issues***
- Fog analytics requires
 - ***Standardization of device and data interfaces***
 - ***Integration with the cloud***
 - ***Streaming analytics to handle continuous incoming data***
 - ***Flexible network architecture where real-time data processing functions move to the edge.***

- Less time-sensitive data can still go to the cloud for long-term storage and historical analysis.
- **Tang et al.** proposed an architecture based on the concept of fog for big data analytics in **smart cities**, which is **hierarchical**, **scalable**, and **distributed**, and supports the integration of a massive number of things and services.
- The architecture consists of **four** layers,
 - **Layer-4** with numerous sensors is at the edge of the network
 - **Layer-3** processes the raw data with many high-performance and low-power nodes.
 - **Layer-2** identifies potential hazards with intermediate computing nodes
 - **Layer-1** represents the cloud that provides global monitoring and centralized control.



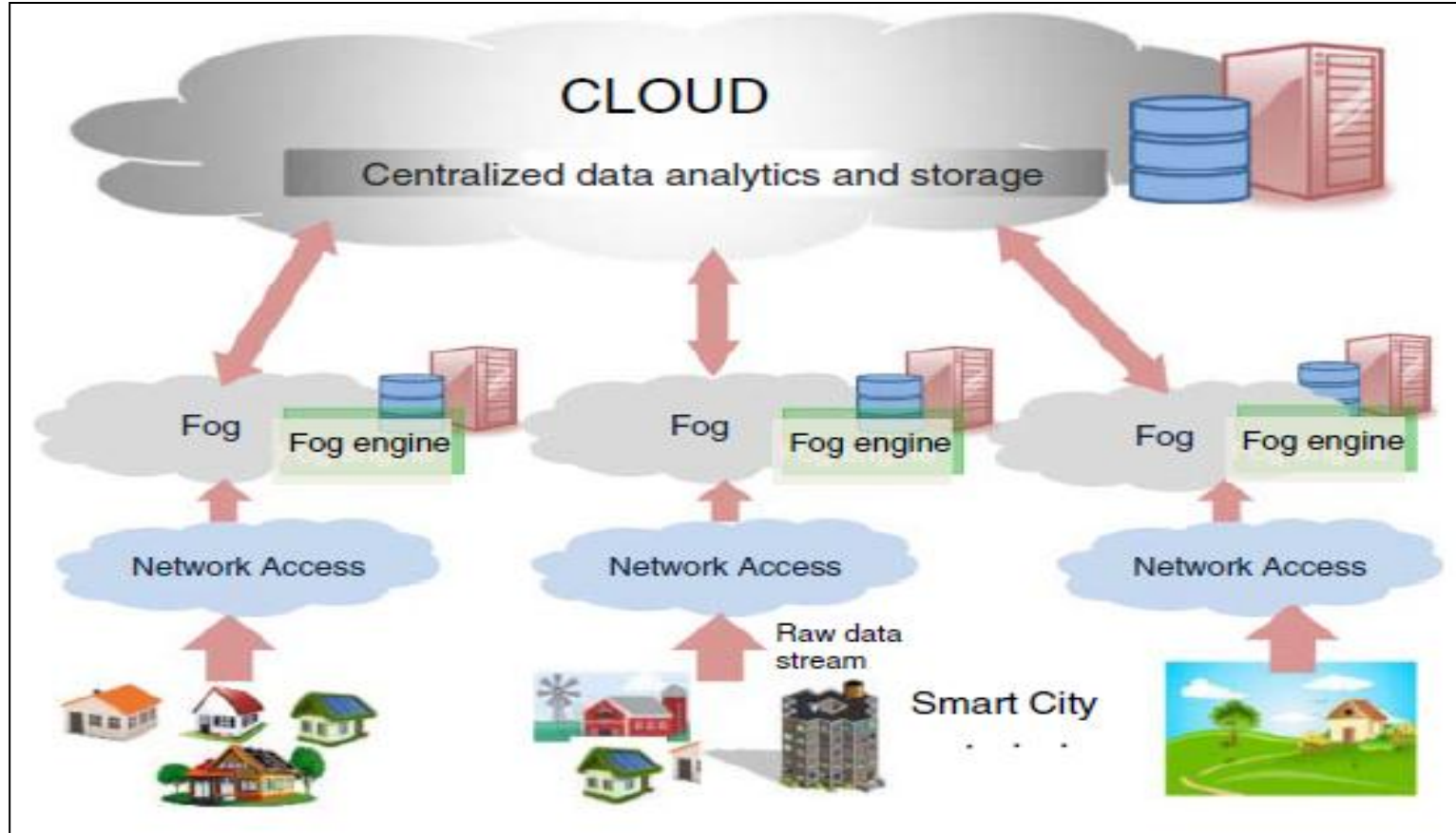
- **FogGIS** a framework based on the fog computing was introduced for *mining analytics from geospatial data*.
- FogGIS performed preliminary analysis, such as compression and overlay analysis, reducing the data sent to the cloud using compression techniques.



- Fog computing is becoming popular in healthcare as more medical devices connect to health IT systems.

ii) Fog-Engines

- Fog-Engine (FE) is an *end-to-end solution* that offers *on-site data analytics* and allows IoT devices to communicate with each other and the cloud.

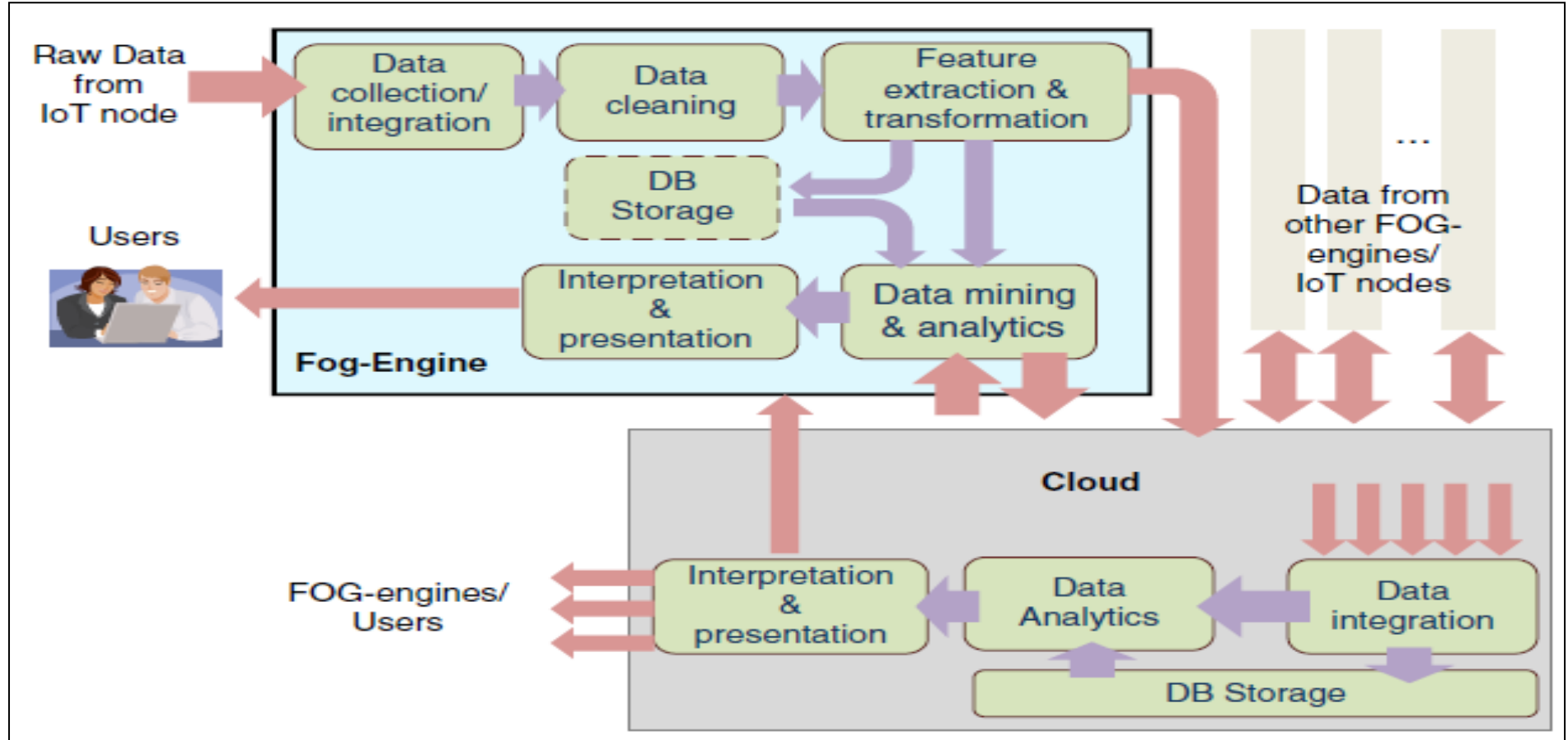


Deployment of FE in a typical cloud-based Computing System

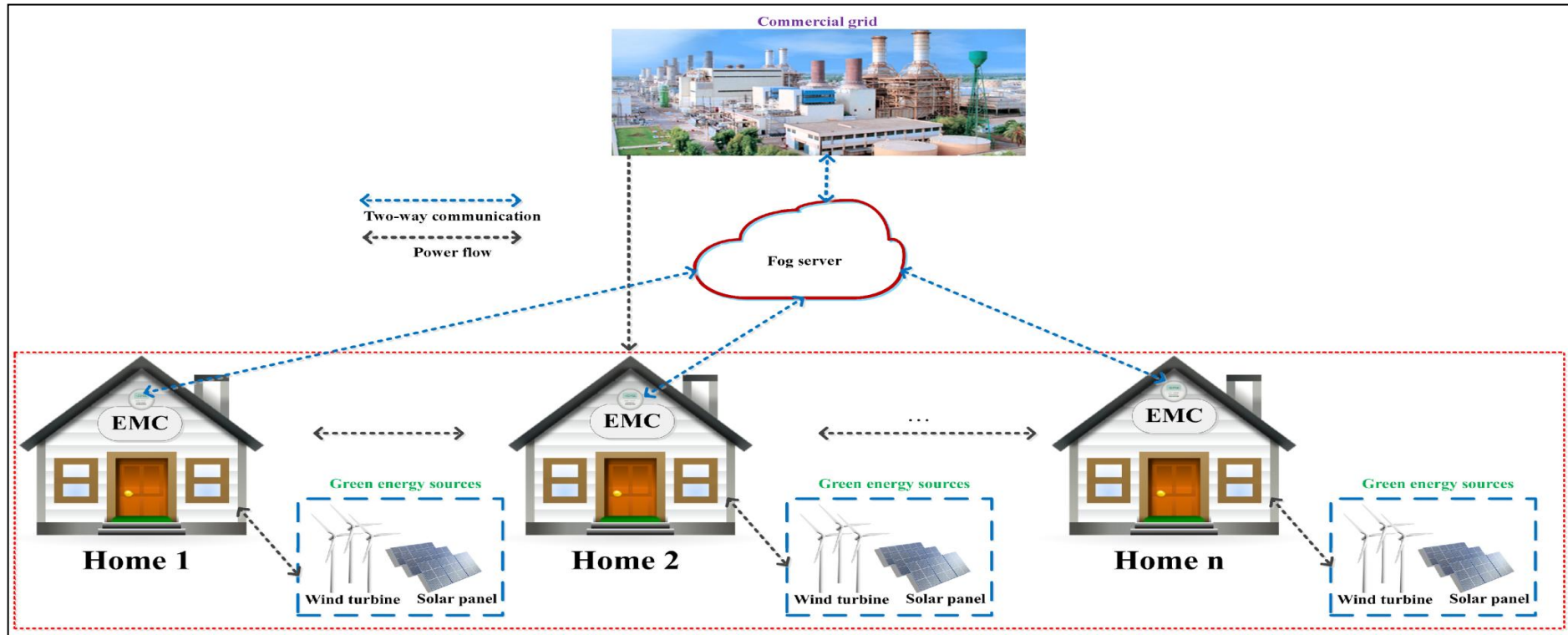
- The fog-engine is a flexible platform used with IoT devices for ***on-premise and real-time data analytics***.
- It allows data processing both in the cloud and at the network edge using connected IoT devices.
- Key features of the fog-engine:
 - ✓ It ***collaborates with nearby fog-engines***, forming a local peer-to-peer network below the cloud.
 - ✓ It ***acts as a gateway***, enabling devices without direct internet access to connect to cloud services.
 - ✓ It ***includes modular APIs*** that offer functions like
 - ✓ **Data offloading**
 - ✓ **Cloud interaction**
 - ✓ **Network management.**
 - ✓ All fog-engines use the same API, ensuring seamless integration with the cloud for IoT developers.

iii) Data Analytics Using Fog-Engines

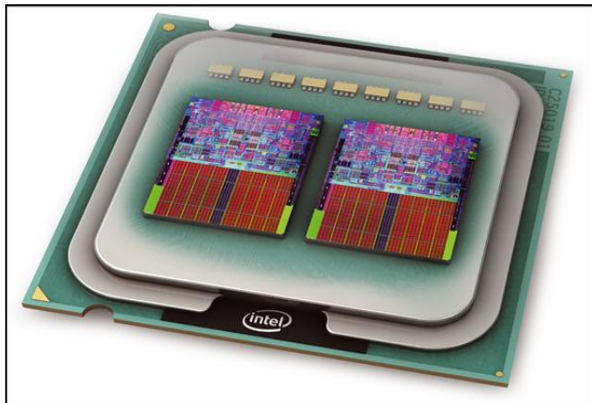
- **On-premise data analytics** being performed near the data source using fog-engines before the data volume grows significantly is shown in the figure



- In a **smart grid**, data is analyzed locally by the Fog Engine (FE) to help users make efficient energy decisions.
- The FE data is then sent to the cloud for large-scale analysis.
- For example, while the FE helps an individual optimize their energy use, the cloud analyzes data from thousands of consumers to help the energy company create better policies.



- Fog engines **update their analytics models** based on policies from the cloud.
- They preprocess, filter, and clean data before sending it to the cloud, reducing the amount of transmitted data.
- Fog engines **perform real-time analytics**, while cloud analytics is offline.
- Fog engines **have less computing power and storage** than the cloud, but cloud processing causes higher latency.
- Fog engines **offer high fault tolerance** by transferring tasks to nearby fog engines during failures.
- A fog engine can use different types of hardware, such as multi-core processors, FPGAs, or GPUs, for more precise processing.
- In contrast, the cloud uses a cluster of similar nodes for large-scale processing.



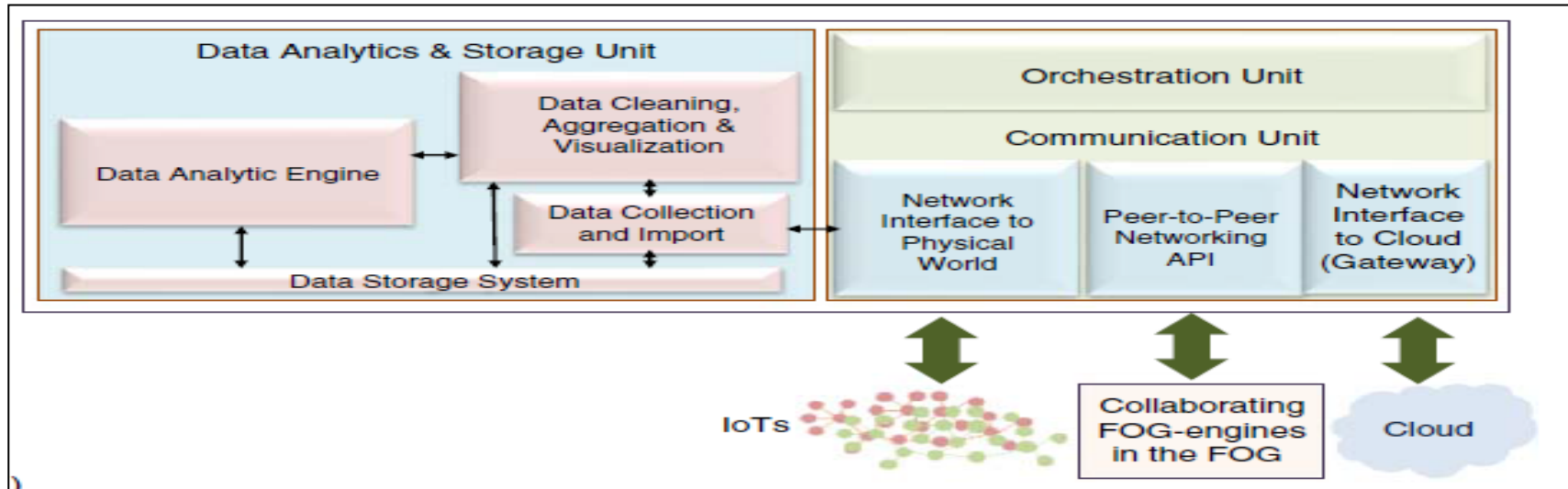
- Fog engines *use fixed hardware* that users can configure, while cloud resources are virtual and beyond user control.
- Fog engines can connect to mobile IoT devices like cars in intelligent transportation systems (ITS).
- Cloud follows a pay-as-you-go model, whereas fog engines are owned by the user.
- Fog engines may run on batteries and need to be energy-efficient, while the cloud has a constant power supply.

- Comparison of fog-engines with cloud computing

Characteristic	Fog-engine	Cloud
Processing hierarchy	Local data analytics	Global data analytics
Processing fashion	In-stream processing	Batch processing
Computing power	GFLOPS	TFLOPS
Network Latency	Milliseconds	Seconds
Data storage	Gigabytes	Infinite
Data lifetime	Hours/Days	Infinite
Fault-tolerance	High	High
Processing resources and granularity	Heterogeneous (e.g., CPU, FPGA, GPU) and Fine-grained	Homogeneous (Data center) and coarse-grained
Versatility	Only exists on demand	Intangible servers
Provisioning	Limited by the number of fog-engines in the vicinity	Infinite, with latency
Mobility of nodes	Maybe mobile (e.g. in the car)	None
Cost Model	Pay once	Pay-as-you-go
Power model	Battery-powered/ Electricity	Electricity

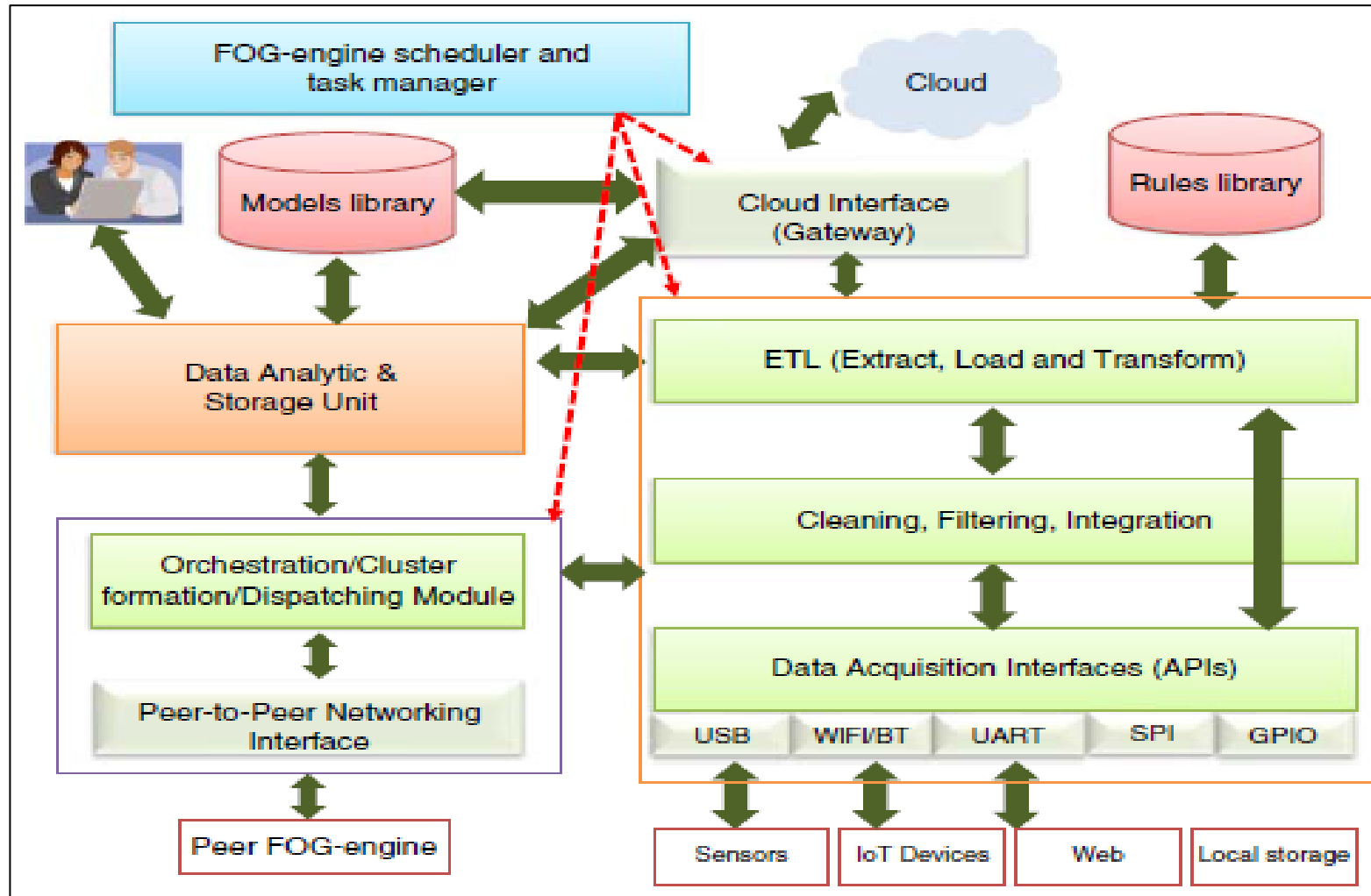
ARCHITECTURE

- The fog-engine is composed of three units:
 1. An **analytics and storage unit** for **preprocessing data, data analytics** and **data storage**
 2. A **networking and communication unit** consisting of the **network interfaces for peer-to-peer networking** and **communication to the cloud and the IoT.**
 3. An **orchestrating unit** to **keep fog-engines synchronized** with each other and the cloud.



General architecture of the fog-engine

- Detailed architecture for the communication unit.



- It uses several common interfaces for acquiring data through
 - *Universal Serial Bus (USB)*
 - *Wi-Fi for mid-range, and Bluetooth for small-range communication with other devices*
 - *Universal Asynchronous Receiver/Transmitter (UART)*
 - *Serial Peripheral Interface (SPI) bus,*
 - *General-purpose input/output (GPIO) pins.*
- The data may be obtained from sensor devices, other IoT devices, web, or local storage.
- The **raw** or **semi structured data** go through preprocessing units such as cleaning, filtering, and integration as well as extract, transform and load, (ETL).
- A **library** keeps the rules, which are **used for data manipulation**.
- The preprocessed data can be transmitted or interchanged with a peer engine via peer-to-peer networking interface unit.

- In a cluster of fog-engines, one with higher processing capacity may act as a cluster head onto which other fog-engines offload the data.
- The **orchestrating unit** handles **cluster formation** and **data distribution across a cluster of fog-engines**.
- The **cloud interface module** is a gateway that facilitates **communication between the fog-engine and cloud**.
- The **FE scheduler** and **task manager** moderate all the above-mentioned units.